X09 WASTE MANAGEMENT | O (MAX : 1 PT)

Intent : Mitigate environmental contamination and associated exposure to hazards present in certain wastes.

Summary : This WELL feature requires the safe management and minimization of wastes associated with hazardous chemicals present in commonly used products.

Issue : Some products may create hazardous waste if handled, transported or disposed of in an uncontrolled manner, increasing risks of exposure to contaminants released to the environment. For instance, unmanaged products containing mercury and other heavy metals may expose people to elevated toxic metals through soil, air and water.[1,2] Leftover pesticides that have become obsolete or otherwise unusable may be disposed in general-purpose dumps, and improper disposal can result in physical injury, environmental pollution and land degradation.[3] Finally, electronic waste, if not properly handled, may create significant health effects downstream. [4]

Solutions : A protocol for handling and minimizing hazardous wastes, which involves separation of hazardous from other solid wastes and procuring adequate receptors for recycling or final disposal, can help mitigate chemical pollution and associated health concerns. By raising awareness and properly managing hazardous wastes, as well as by selecting products that are easier to reuse and have a lower impact on human health, projects may minimize the generation of such wastes and the release of hazardous materials into the environment.

PART 1 IMPLEMENT A WASTE MANAGEMENT PLAN (MAX : 1 PT)

For All Spaces:

For all batteries, pesticides, lamps that may contain mercury, other mercury-containing equipment (including thermostats and thermometers),[5] and electrical and electronic equipment[6] present or expected to be present within the project during the building operations, a waste management plan that contains the following is developed and implemented:

a. Identification of roles, responsibilities and vendors for implementing the plan.[7]
b. Identification of the sources of waste, estimation of rates of generation and strategies to minimize waste generation.[7]
c. Strategies for waste collection. Each of the categorized wastes is separately contained in clearly labeled receptacles and removed from the building within one year.[5]
d. Protocols for cleaning spills of mercury (including broken fluorescent lamp tubes), pesticides and battery electrolyte fluid, including sealed containment of residues, as applicable.[5]
e. Protocols to track, measure and report waste stream flows.[7]
f. Protocols for off-site shipment of wastes.

REFERENCES

1. UNEP DTIE Chemicals Branch World Health O. Guidance for Identifying Populations at Risk From Mercury Exposure. 2008.
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4. Grant K, Goldizen FC, Sly PD, et al. Health consequences of exposure to e-waste: a systematic review. The Lancet Global Health. 2013;1(6):e350-e361.
5. U.S. Environmental Protection Agency. Standards for Universal Waste Management. In: U.S. Environmental Protection Agency, ed. Vol 40 CFR 2732012.
6. European Parliament and the Council of the European Union. Directive 2012/19/EU of the European Parliament and of the Council of 4 July 2012 on waste electrical and electronic equipment (WEEE). In: Union E, ed. 2012/19/EU. Vol 2012/19/EU2012.
7. Bill Conley SJ. Sustainability How-to Guide – Waste Stream Management. International Facility Management Association;2016.